

Mnemosyne — Master Execution Plan

Goal: the world's best-benchmarked AI memory system + the field's neutral benchmark leaderboard

Version: 1.0 · **Date:** 2026-07-08 · **Status:** Proposed (awaiting go) **Owner surface:** Mnemosyne core ([src/mnemosyne/](#) , [eval/](#)) + new greenfield [leaderboard/](#) and [web/](#) . **Audience:** an autonomous engineering agent (or agent fleet) executing end-to-end, plus human operators for gated evidence capture.

0. How to use this document (executing agent, read first)

This is an execution spec, not prose. Work top-to-bottom within a workstream; respect the dependency map (§8). Every task has a stable ID ([A1.3](#)), a **Definition of Done (DoD)**, and required **evidence artifacts**. Nothing is "done" without its artifacts.

Hard guardrails (violating any = stop and escalate):

- 1. **Do not break the §31 invariant rails or the 5 §33 test classes.** All existing [eval/](#) gates and [tests/](#) must stay green. Memory-capability work is additive.
- 2. **Honesty-charter compliance is reframed, not relaxed (see §2).** Private-suite numbers stay internal. Public claims are permitted *only* on public benchmarks, *only* with the full reproducibility bundle (§A5), and *only* with retrieval-recall and LLM-judged-QA reported as separate columns. No blended "we're #1" claims.
- 3. **Never tune on a held-out/test split.** Contamination discipline (§B1.C) applies to our own system exactly as to submitters.
- 4. **No self-defined benchmark may be a headline claim.** (This is the gbrain "BrainBench" anti-pattern; see §6.) Private suites remain internal QA only.
- 5. **All perf/scale claims require measured evidence**, per the existing §11 "measurement-gap register." No asserted-but-unmeasured numbers ship.

Ground-truth references in-repo (source of truth over this doc if they conflict): [docs/ARCHITECTURE-OVERVIEW.md](#) , [docs/ENGINE-CONTRACT.md](#) , [docs/blueprint/Mnemosyne-Performance-and-Refactoring-Blueprint.md](#) (Waves A–E, §9.2.7, §9.7, §10), [docs/ROADMAP-T0-100.md](#) , [.planning/STATE.md](#) , [eval/README.md](#) , [docs/research/AI-Memory-Systems-Market-Research-2026.md](#) .

DoD template: *Code merged + tests green + artifacts written to the named path + a one-paragraph result note in [eval/reports/](#) or [leaderboard/reports/](#) + CI gate updated.*

1. Objectives & success criteria (measurable)

Goal A — Best-benchmarked memory system

#	Win condition	Metric	Target
A-i	Top-tier deterministic retrieval on a public suite	LongMemEval Recall@5 (retrieval track, no LLM in scoring)	≥ 97.6% (matches the best published no-LLM score) with full artifacts
A-ii	Competitive multi-hop retrieval	HippoRAG suite Recall@2 / @5 on MuSiQue / 2Wiki / HotpotQA, EM/F1	Meet/beat published HippoRAG 2 baselines
A-iii	Close the acknowledged QA gap	LongMemEval / BEAM QA accuracy (LLM-judged, disclosed reader)	From current private qa_hard_v2 ≈ 0.62 → ≥ 0.85 on public QA

#	Win condition	Metric	Target
A-iv	Own the empty columns	Poison-block / attack-success-under-defense; calibration ECE vs public labels	Publish both; be the only system that does
A-v	Credible at scale	Warm and concurrent P95; 100k-item cells	Measured distributions, not point claims

Goal B — The neutral leaderboard

#	Win condition	Definition
B-i	It exists and is trusted	Public site + open harness + methods paper + openly-licensed results dump
B-ii	It is <i>structurally</i> neutral	Operator runs every system under one harness; hard firewall on our own entry (\$2, \$B0)
B-iii	It resists gaming	Hidden/rotated test split, contamination controls, signed build==public attestation, honest LLM-judge reporting
B-iii	It becomes the default	Multi-track (conversational / agentic / multi-session); all major systems present; academic co-sign

Explicit non-goals / anti-patterns to avoid

- No headlining **LoCoMo** or **DMR/MS**C (contested / saturated — see market research §6).
- No presenting **recall@k as if it were QA accuracy** (the MemPalace failure).
- No `top_k = entire-candidate-pool` "100%" retrieval theater.
- No "neutral" leaderboard that quietly advantages our own entry (the "Leaderboard Illusion" failure).

2. Strategic framing — the two hard problems, resolved

Problem 1 — the honesty-charter tension. The perf blueprint §9.7 and `eval/provider_bakeoff/README.md` currently forbid citing public benchmarks in headline claims. That rule was correct *while no public benchmark had been run under our discipline*. The resolution is not to relax it but to **upgrade it into a Public-Benchmark Publication Protocol (PBPP)** that is stricter than what any competitor does:

PBPP: A public number may be published only if (a) it was produced by the pinned public harness in `eval/public/`, (b) the full artifact bundle (§A5) is released simultaneously, (c) retrieval-recall and LLM-judged-QA are reported in separate columns with the judge model + prompt disclosed, (d) the private golden suite is never conflated with it, and (e) an independent third party can reproduce it from the bundle. Private-suite numbers (0.977 / 0.983 / ECE 0.0063 / poison 1.0 / 149.5 ms) remain internal QA and are never headline public claims.

This *turns honesty into the moat*: our published numbers become the most reproducible in a field full of contested vendor claims. Update `docs/blueprint/...§9.7` and the provider-bakeoff README to reference PBPP rather than a blanket prohibition.

Problem 2 — the neutrality paradox. We cannot be both the referee and the champion on trust alone; the research is unambiguous that vendor-run-where-they-win leaderboards get dismissed (OmniMemEval, omegamax, the LMArena "Leaderboard Illusion"). Resolution (§B0): the leaderboard runs under **independent governance with a hard firewall**, the operator **runs every system itself under one identical harness** (killing the "you misconfigured us" defense that defined the mem0 ↔ Zep dispute), Mnemosyne is entered and scored **by the same rules as everyone else**, and all raw artifacts are public. We win by having the best *reproducible* numbers on someone else's neutral turf — not by controlling the scoreboard.

3. The eclectic architecture synthesis (best ideas → mapped to what Mnemosyne already has)

Mnemosyne already implements a remarkable share of the field's best ideas. This table is the design ledger: for each capability, the best external idea, its source, Mnemosyne's current state, and the action. **Priority = benchmark ROI.**

Capability	Best external idea(s)	Source	Mnemosyne today	Action	Prio
Two-tier episodic ↔ semantic	Complementary Learning Systems	McClelland/ Kumaran/ Hassabis	Evidence ledger (episodic) + typed projections (semantic) — already CLS-shaped	Keep ; frame explicitly	—
Associative recall	Graph index + Personalized PageRank	HippoRAG / HippoRAG 2	Have it (<code>algorithms.ppr_power_iteration</code> , <code>graph_ppr_cache</code>)	Extend (HippoRAG2 deeper passage integration)	High
Hierarchical sensemaking	Community summaries	Microsoft GraphRAG	RAPTOR gist tree in <code>summarizer</code>	Keep ; add global map-reduce query mode	Med
Multi-hop answer synthesis	Iterative retrieve → read; reader model	HippoRAG2, LongMemEval readers	Weak spot (<code>qa_hard_v2</code> ≈ 0.62)	Build — the #1 lever (\$A1)	Top
Consolidation loop	Offline replay + distillation ("sleep/ dreaming")	Google "Sleep", OpenAI Dreaming	11-role warm loop (<code>replayer...</code> <code>user_model_updater</code>)	Keep ; add async batch "sleep" job	Med
Multi-timescale memory	Continuum Memory System (freq-staggered modules)	Google Nested Learning / Hope	Single warm loop cadence	Extend — spectrum of update rates → anti-forgetting	High
Write gating	Surprise (gradient magnitude)	Titans; EM-LLM (Bayesian surprise)	Promotion gate, mutation budget	Extend — surprise signal into <code>promotion_gate</code>	Med
Episodic segmentation	Event boundaries + temporal-contiguity retrieval	EM-LLM	Bitemporal, but no event-segmented recall	Add — episode boundaries + temporal-adjacency reads	High
Forgetting / deletion	One-shot selective forget; adaptive decay	Larimar; Titans; ACT-R	Fidelity tiers + crypto-shred + optional ACT-R decay	Keep (already best-in-class)	—
Durable parametric facts	Localized weight edits; sparse KV memory layers	MEMIT; Meta Memory Layers	Parametric tier (LoRA/TTT, <code>parametric.py</code>)	Extend — evaluate MEMIT-style edits as a write path	Low
Compiled context	Offline-trained reusable KV cache	Cartridges (Stanford)	None	Research — per-tenant "cartridge" for hot corpora (latency/ throughput win)	Med
Retrieval/param split	Small model + huge swappable DB	RETRO	Already retrieval-first over Postgres	Keep	—
Bounded working memory	Compressive per-layer memory; memory tokens	Infini-attention; RMT	N/A (model-agnostic)	Skip (out of scope; model-internal)	—
Activation-space memory + introspection	J-space / global workspace, persona vectors, SAE features, concept injection	Anthropic (§3.1)	None	Research track (§3.1) — calibration + security differentiator	Med

Capability	Best external idea(s)	Source	Mnemosyne today	Action	Prio
Agent memory API contract	<code>/memories</code> file ops + context editing + compaction	Anthropic memory tool / context engineering	<code>mneme</code> CLI + 48 MCP tools	Align the MCP memory surface to this contract	Med

3.1 The Anthropic introspection / "J-space" track (novel differentiator)

Anthropic's 2025–26 interpretability line is directly usable — with strict epistemic humility (we make **functional** claims, never consciousness claims). Requires **white-box access to open-weight models**, which we have via the self-hosted Ollama role-LLMs (`infra` Ollama) — so this is feasible on our own stack, not on closed APIs.

- **"A global workspace in language models" / J-lens** (Anthropic, Jul 2026; `github.com/anthropics/jacobian-lens`): a readable list of "words on the model's mind but not said." **Use:** log a compact J-lens readout as an activation-trace alongside a memory write; use it as a **poisoning/deception tripwire** (evaluation-awareness, fabrication markers) on the consolidation path.
- **"Emergent introspective awareness"** (Lindsey, Oct 2025) via **concept injection**: self-reports are a **weak (~20% reliable), calibratable** signal. **Use:** feed gated introspective confidence into the existing conformal-calibration/abstain layer — never as sole ground truth.
- **Persona vectors** (Anthropic, Aug 2025; arXiv 2507.21509): linear activation directions for traits, extracted by contrastive prompting. **Use:** rolling projection = a **drift-detection** signal for the user-model/consolidation loop; screen candidate memories/adaptor data before persistence.
- **SAE features** (Scaling Monosemanticity): a stable, inspectable concept vocabulary. **Use:** tag memories by active interpretable features → activation-space retrieval + audit.
- **Dual-use warning:** concept injection *is* memory poisoning (the "bread" false-intention experiment). Any activation-level memory write path must carry provenance + capability checks exactly like text writes (§31 rails).

DoD for the track: a `research/activation-memory/` spike with (1) J-lens readouts wired as an optional consolidation tripwire on the Ollama role-LLM, (2) a persona-vector drift metric in the ops dashboard, (3) a written go/no-go on promoting any of it to core. Explicitly labeled research; not on the critical path to Goal A.

4. WORKSTREAM A — Make Mnemosyne the best-benchmarked system

A0 — Wire public benchmarks into `eval/public/` (deterministic-first) — *this is Wave E kickoff*

Blueprint §9.2.7 already slates "LongMemEval/BEAM internal gate wired into `eval/` ." Build it as a new isolated tree so the private suite stays clean.

- **A0.1 Harness scaffold.** Create `eval/public/` with a runner that drives *only the public* `mneme` CLI (same rule as `eval/run_eval.py`), pins each benchmark to a commit, and emits the §A5 artifact bundle. DoD: `mneme eval-public --suite X` runs end-to-end and writes traces.
- **A0.2 LongMemEval — retrieval-recall track first.** Deterministic Recall@k / nDCG using the dataset's session/turn gold labels; **no LLM in scoring**. DoD: R@5 reported with Wilson CI + per-question retrieval traces. Target A-i.
- **A0.3 HippoRAG multi-hop suite.** MuSiQue / 2WikiMultiHopQA / HotpotQA via recall@2/@5 + EM/F1 (fully deterministic). This exercises the graph+PPR channel we already have. DoD: table vs published HippoRAG 2 baselines. Target A-ii.
- **A0.4 MemoryAgentBench adapter (upstream PR).** Mostly deterministic (SubEM/exact-match); contribute a Mnemosyne adapter to the upstream repo — the closest thing to an "official submission." DoD: adapter merged or PR open + local results. Targets conflict-resolution (our TMS/belief-revision home turf).
- **A0.5 BEAM (prestige, LLM-judged).** Run with a disclosed reader model; publish as "Mnemosyne + ." DoD: BEAM-1M result + full config disclosure. Target A-iii.

- **A0.6 Gate wiring.** Add a non-blocking CI job that runs the deterministic public suites on a schedule; regression alerts only (never tune-to-test).

A1 — Close the multi-hop answer-synthesis gap (the top lever)

Retrieval is already strong; synthesis (~0.62) is the ceiling on QA benchmarks. - **A1.1** Add an **iterative retrieve → read** loop over the existing hybrid retriever (query decomposition → PPR spreading-activation hops → evidence assembly). Borrow HippoRAG 2's deeper passage integration and EM-LLM's temporal-contiguity read. - **A1.2** Add a disclosed **reader/synthesis** step (self-hosted role-LLM) that composes an answer *only* from retrieved, provenance-tagged evidence (fail-closed as data, per §31 R6). - **A1.3** Surprise-gated + event-segmented episodic recall (Titans/EM-LLM) to pull coherent episodes, not fragments. - **DoD:** `qa_hard_v2` and LongMemEval QA ≥ 0.85; per-hop retrieval traces prove the answer is grounded. Target A-iii.

A2 — Capability upgrades (eclectic adds), by benchmark ROI

- **A2.1 Multi-timescale consolidation spectrum** (Nested Learning CMS): parameterize the warm loop into fast/medium/slow cadences so new writes don't overwrite semantic structure. DoD: continual-learning eval shows reduced forgetting.
- **A2.2 Async "sleep" consolidation job** (Google Sleep / OpenAI Dreaming): offline replay + distillation of durable facts into slower tiers + **temporal freshness/auto-expiry** (facts age correctly). DoD: staleness metric added; freshness regression test.
- **A2.3 Global map-reduce query mode** (GraphRAG) over the RAPTOR tree for "themes across everything" questions. DoD: sensemaking eval cell.
- **A2.4 (Research) Cartridges** compiled per-tenant KV cache for hot corpora. DoD: latency/throughput A/B vs retrieval baseline.

A3 — Security & calibration as *published* differentiators (fills the empty columns)

- **A3.1** Expand the 59-attack poison suite (`poison_suite.json`) to the full **MINJA + AgentPoison + PoisonedRAG** threat models; publish **attack-success-under-defense**, not just block-rate. DoD: adversarial report with ASR curves.
- **A3.2** Publish **calibration curves + ECE against public benchmark labels** (not only the private set). DoD: reliability diagram in the artifact bundle.
- **A3.3 (from §3.1)** Optional J-lens / persona-vector tripwires as an *additional* published robustness signal.
- **Why:** in the market survey, *no competitor publishes either column*. This is a near-uncontested win.

A4 — Performance / scale close-out (Phase 9 / Waves B–D) so results are credible

Clear the §11 measurement-gap register: provider bake-off + default (TEI vs sidecar vs ONNX), null-embedding backfill evidence, halfvec/pgvectorscale DiskANN, 100k-item cells, and **concurrent** P95 (not just warm-serial 149.5 ms). DoD: measured distributions in `eval/latency*` + `eval/provider_bakeoff/`.

A5 — The reproducibility artifact bundle (PBPP standard)

Every public result ships: pinned harness commit + `pip / uv` runner; per-question traces (**what was stored, what was retrieved, final answer**); disclosed judge model + prompt + all configs; system build fingerprint (we already emit `sha256:...` release fingerprints); Wilson/bootstrap CIs; a one-command reproduce script. **DoD:** an independent agent reproduces the number from the bundle alone. Then commission a genuine third-party reproduction before any public claim.

Workstream A exit gate: A-i...A-v met, PBPP satisfied, third-party reproduction on file, honesty-charter docs updated to PBPP.

5. WORKSTREAM B — The neutral Memory Leaderboard + website

Working name: **OpenMemBench** (final naming = branding decision; recommendation: a neutral, non-Mnemosyne-branded name to preserve independence).

B0 — Governance & neutrality FIRST (this is the credibility moat, build before code)

- **B0.1 Independent governance charter.** Multi-institution advisory board (invite ≥ 2 academics who authored memory benchmarks); published, dated COI policy; the operator's own entry (Mnemosyne) gets **no** extra runs, no private tuning window, no earlier held-out access, no score retraction rights. Model on HELM's disclosed-funding posture + LMArena's post-"Illusion" reforms.
- **B0.2 One-harness rule.** Operator runs **every** system itself under one identical harness; vendor-submitted numbers are never accepted as-is. This structurally eliminates the mem0 $\leftrightarrow$ Zep "you misconfigured us" dispute.
- **B0.3 Survivability.** Openly-licensed results dump (durable, citable URLs) + non-single-owner hosting so the board outlives its host (the Papers-with-Code lesson).
- **B0.4 Methodology paper.** Publish a peer-review-grade methods paper before/at launch (the arXiv 2603.07670 "GLUE-style shared leaderboard" gap is the explicit opening).

B1 — Benchmark methodology & harness (the product)

Synthesized requirements checklist (each maps to a proven precedent): - **B1.a Multi-track, GLUE-style:** conversational · agentic · multi-session tracks, shared metric stack. - **B1.b Separate metric families:** deterministic retrieval (recall@k, EM, programmatic assertions) vs LLM-judged QA — **never blended** (the core field failure). - **B1.c Contamination controls:** mandatory machine-validated training-data disclosure (MTEB [training_datasets](#)); surface a **zero-shot overlap** score *without hiding honest disclosers*; fresh-vs-circulated probes (SWE-bench-Illusion method). - **B1.d Hidden/held-out + rotation:** publish train/dev, keep a **private scoring split** (ARC-AGI); public/private split with periodic reveal (Kaggle); scheduled task rotation + human baseline per track (GLUE saturation lesson). - **B1.e Anti-gaming:** signed **attestation that the evaluated build == public release** (LMArena's Llama-4 fix); submission throttling (SuperGLUE 2/day); disclose *all* variants tested, not just the winner; public retirement list. - **B1.f Keep the judge honest:** publish the judge's acceptance rate on intentionally-wrong-but-topical answers (beat LoCoMo's 62.8%); multi-judge averaging; rotating human adjudication; prefer deterministic grading wherever possible. - **B1.g Systems-fair reporting:** efficiency (latency + cost/query) as first-class, budget-normalized; CIs + rank ranges with overlapping-CI systems treated as **ties**; set-independent aggregation (Borda, not mean-win-rate). - **B1.h Reproducibility:** pinned harness commit, full per-question traces public, open-source-to-rank.

B2 — Website architecture & build (greenfield [web/](#) + [leaderboard/](#))

- **Stack (recommendation):** static-first for durability + trust — **Astro or Next.js (static export)** front-end; results as **flat, versioned JSON/Parquet in a public git repo** (not a hidden DB) so every number is auditable and URLs are permanent; charts via a light lib; a **per-question trace browser** (the HELM "browse all predictions" pattern). Host on a CDN; mirror the data dump.
- **Data model:** `system → track → benchmark_version → run(commit, build_fingerprint, config, judge) → per_question_traces → metrics(with CIs)`.
- **Submission flow:** reviewed **PR with provenance link** (MTEB model); CI validates schema + attestation; organizer runs the eval; results + traces published.
- **DoD:** site renders the leaderboard from the public data repo; one system fully browsable end-to-end (traces included).

B3 — Content / explainer layer ("explain memory to everyone")

- Plain-language explainers of each major system's architecture (reuse [docs/research/AI-Memory-Systems-Market-Research-2026.md](#)), the **recall-vs-QA** distinction front-and-center, and a transparent methodology page. DoD: explainer pages published; reviewed for accuracy against primary sources.

B4 — Launch & community strategy

Seed the board by running **all major systems** (mem0, Zep/Graphiti, Letta, Cognee, MemOS, supermemory, HippoRAG, +Mnemosyne) under the one harness; publish the methods paper; open submissions + a public audit/dispute channel; **release raw head-to-head data** (LMArena's single most trust-restoring act). DoD: public launch with ≥8 systems and a reproducible methods paper.

Workstream B exit gate: B-i...B-iv met; independent board seated; ≥8 systems live; methods paper public; Mnemosyne entered under identical rules.

6. Cross-cutting: anti-gaming, verification, risks

Failure modes to design against (from real 2026 episodes): - MemPalace: recall@k reported as QA; `top_k=whole pool`; hand-patching dev questions ("teaching to the test"). → Enforced by B1.b, B1.d, and per-question trace publication. - gbrain: self-defined "BrainBench" headline. → Forbidden by guardrail #4 / PBPP. - LMArena "Leaderboard Illusion": operator/unequal access. → Neutralized by B0.1–B0.2. - LoCoMo: 6.4% wrong answer key + lenient judge. → We don't headline LoCoMo; our judge honesty is published (B1.f).

Verification requirements: every Goal-A public number reproduced by an independent party before publication; every leaderboard result reproducible from its bundle; a standing "red-team the number" review before any external claim.

Risk register (top items): | Risk | Mitigation | |---|---| | Neutrality perception (we compete + operate) | Independent board, one-harness rule, no special access, raw-data release (§B0) | | Honesty-charter breach | PBPP (§2); private suite never headlined | | Multi-hop QA underperforms | A1 is top priority + gated; if <0.85, publish deterministic-retrieval wins only | | Benchmark saturation/contamination | Hidden split + rotation + fresh scenarios (B1.d) | | Compute cost of running all systems | Budget realistically; curated reference set + community queue (Open-LLM-Leaderboard lesson) | | Over-claiming | Recall vs QA separated; CIs + ties; "provisional" until reproduced |

7. Master timeline, roles, dependency map

Sequencing (critical path in bold): 1. **§2 charter** → **PBPP** (unblocks all publication) — days. 2. **A0 public-benchmark harness** (Wave E kickoff) || **B0 governance charter** — weeks 1–3. 3. **A1 multi-hop synthesis** (top lever) || **B1 methodology + B2 site scaffold** — weeks 2–8. 4. A2 capability upgrades || A3 security/calibration publish || B3 explainers — weeks 4–10. 5. **A4 perf/scale close-out + A5 third-party reproduction** — weeks 6–12. 6. **B4 launch** (seed ≥8 systems, methods paper) — weeks 8–14. 7. A3.1/§3.1 research tracks — parallel, off critical path.

Dependencies: A0 blocks A1/A3.2/A5; §2 blocks any publication; B0 blocks B4; A0's harness is reused by B1 (shared eval core).

Roles: *core-eng agent* (A0–A2 code), *ml agent* (A1 reader, §3.1 research), *sec agent* (A3), *infra agent* (A4), *web agent* (B2–B3), *governance/human operator* (B0 board, A5 third-party repro, evidence capture). Run independent workstreams as parallel sub-agents; serialize shared-eval-core writes.

Consolidated Definition-of-Done checklist

- [] Charter updated to PBPP; provider-bakeoff README references it.
- [] `eval/public/` harness live; LongMemEval-recall, HippoRAG multi-hop, MemoryAgentBench adapter, BEAM all runnable with artifact bundles.
- [] Multi-hop QA ≥ 0.85 with grounded per-hop traces.
- [] Poison ASR-under-defense + calibration-vs-public-labels published.
- [] Concurrent + warm P95 and 100k-cell numbers measured.
- [] Independent third-party reproduction of headline public numbers on file.

- [] Governance board seated; COI + firewall policy public.
- [] Leaderboard site live from a public, versioned results repo; ≥8 systems; trace browser working.
- [] Methods paper published; raw data dump openly licensed.
- [] Mnemosyne entered under identical rules; no special access.

8. Appendix A — Recommended benchmark slate (deterministic-first)

Tier	Benchmark	Track	Scoring	Why
Primary	LongMemEval (retrieval-recall)	conversational	Deterministic	Publishable recall@k; the flag everyone plants
Primary	HippoRAG suite (MuSiQue/2Wiki/HotpotQA)	multi-hop	Deterministic (R@k, EM/F1)	Cleanest deterministic; exercises our PPR channel
Strong	MemoryAgentBench	agentic/conflict	Mostly deterministic	Upstream adapter = closest to an "official" submission
Prestige	BEAM (1M/10M tok)	long-horizon	LLM-judged (disclosed reader)	Un-saturated; differentiating at scale
Utility	STATE-Bench (MS)	enterprise task success	Deterministic task-completion	Shows memory improves real tasks
Internal-only	LongMemEval-QA, private v2 suite	—	LLM-judged / private	QA signal + regression; never a headline claim
Avoid headlining	LoCoMo, DMR/MS	—	contested/saturated	Run silently at most; never lead with them

Appendix B — Source map

- **Frontier-lab architecture:** Titans (2501.00663), Infini-attention (2404.07143), Nested Learning/Hope (2512.24695), Google "Sleep" (2606.03979), Memorizing Transformers (2203.08913), RETRO (2112.04426), Meta Memory Layers (2412.09764), Larimar (2403.11901), GraphRAG (2404.16130), MEMIT (2210.07229), HippoRAG/2 (2405.14831 / 2502.14802), EM-LLM (2407.09450), TTT (2407.04620), Cartridges (2506.06266), CLS (Kumaran/Hassabis/McClelland 2016).
- **Anthropic:** global workspace / J-lens (transformer-circuits.pub/2026/workspace; github.com/anthropics/jacobian-lens), introspective awareness (transformer-circuits.pub/2025/introspection), persona vectors (2507.21509), monosemanticity (2024 scaling), memory tool + context engineering (platform.claude.com docs; anthropic.com/engineering/effective-context-engineering-for-ai-agents).
- **Leaderboard credibility:** MTEB (2506.21182), HELM (2211.09110), Chatbot Arena (2403.04132) + Leaderboard Illusion (2504.20879), GLUE/SuperGLUE (1804.07461 / 1905.00537), SWE-bench Verified (openai.com), ARC-AGI (arcprize.org), memory-leaderboard call (2603.07670).
- **In-repo:** [docs/research/AI-Memory-Systems-Market-Research-2026.md](#) , [docs/blueprint/Mnemosyne-Performance-and-Refactoring-Blueprint.md](#) (Waves A–E), [docs/ROADMAP-T0-100.md](#) , [eval/README.md](#) .

This plan is additive to the attested v1.0 system. It does not modify the §31 rails or §33 gates; it wires public benchmarks, closes the multi-hop synthesis gap, publishes the security/calibration columns no competitor fills, and stands up an independently-governed leaderboard on which Mnemosyne competes under the same rules as everyone else.